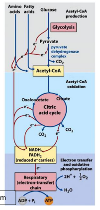
LECTURE 4

See PDH & TCA notes

Catabolism of Proteins, Fats & Carbohydrates: Three stages of Cellular Respiration



1: Oxidation of fatty acids, glucose, and some amino acids yields acetyl-CoA.

2: Oxidation of acetyl groups in the citric acid cycle includes four steps in which electrons are abstracted

3. Electrons carried by NADH and $FADH_2$ are funneled into the respiratory chain and the production of ATP.

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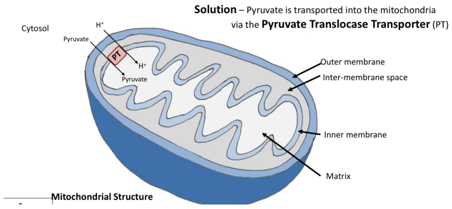
Problem - Pyruvate is generated in the cytosol by Glycolysis
- Enzymes for conversion of pyruvate to Acetyl-CoA exist in the mitochondria

Solution – Pyruvate is transported into the mitochondria via the **Pyruvate Translocase Transporter** (PT)



Mitochondrial Structure

Clinical Notes: PDH Deficiencies

- **Several vitamins required for PDH activity:**
  - Vit $B_5 \rightarrow$ Co-enzyme A,
  - Vit $B_3 \rightarrow$ NAD,
  - Vit $B_2 \rightarrow$ FAD,
  - Vit $B_1 \rightarrow$ TPP
  - Deficiency in any of these vitamins will disrupt PDH activity

Clinical Notes: Thiamine Deficiency Causing Low PDH Activity

- **Wernicke-Korsakoff syndrome** characterized by Ataxia, Ophthalmolplagia, Memory loss, Cerebral Hemorrhage, Confabulation
- At risk: Alcoholics & malnourished individuals
- **Beriberi:** wet and dry based on presence/absence of oedema
  - Wet Beriberi: Heart failure, decrease ATP, increased cardiac output, (dilated cardiomyopathy)
  - Dry Beriberi: systemic muscle wasting, polyneuritis

Other thiamine –requiring enzymes:
- α-ketoglutarate DH (αKG DH) part of the TCA cycle
- Branched chain α–ketoacid DH (BCKDH)

Clinical notes: Pyruvate dehydrogenase deficiency

Metabolic effect:
↑ in pyruvate with concomitant ↑ in lactic acid and alanine (via transamination), ↓ in production of acetyl CoA; severe reduction in ATP production

Clinical features:
lactic acidosis in blood and CSF, increased blood pyruvate and alanine, neurologic defects, myopathy;
usually fatal at early age

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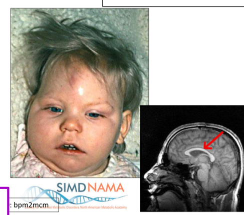
Therapies:
Dichloroacetate inhibits PDH kinase so that PDH complex remains unphosphorylated and catalytically as active as possible
Supplementation with thiamine, lipoic acid and carnitine
High fat and low carbohydrate diet

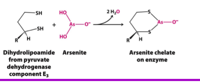
PDH Deficiency



- Frontal prominence
- Wide nasal bridge
- Flared nares
- Long philtrum
- Brain malformations
  - Corpus callosum agenesis
  - Cerebral and basal ganglia cysts

Heavy metal poisoning from the disruption of pyruvate metabolism

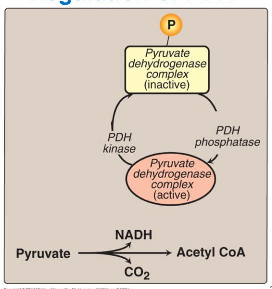
- Arsenate, Mercury, and lead have high affinity for -SH
- Lipoic acid is one of the cofactors in PDC
- PDH Complex becomes inactive when lipoic acid is bound to heavy metals.
- CNS solely depends on Glucose Metabolism through to oxidative phosphorylation for energy, (can not use fatty acids as a fuel source) therefore effected by heavy metal poisoning.



Regulation of PDH

High Energy Charge OFF

Low Energy Charge ON



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Activation of PDH Complex

Activation by phosphatase

- **by dephosphorylation of serine residues on E1 PDH by PDH Phosphatases**

Phosphatase stimulated by:
- Calcium particularly in skeletal muscle during contraction
- Insulin in adipocytes and liver
- Catecholamines in cardiac muscle

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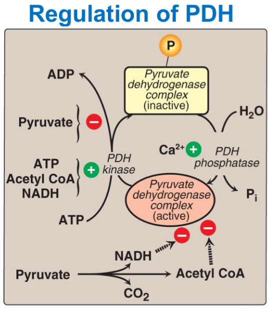
SOM.1c.BPM2.2.DM.3.BCHM.N

Regulation of PDH

High Energy Charge OFF

Low Energy Charge ON



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Skeletal Muscle
Ca^{2+} release during contraction – energy production.

Cardiac Muscle
catecholamines.

Important regulatory molecules – NADH, Acetyl-CoA, Pyruvate, ATP and calcium.

Inhibition of PDH Complex

- Inhibition of PDH by reaction products
  - Acetyl CoA & NADH

- **Inhibition of PDH upon phosphorylation of serine residues on E1 by PDH kinases**

Kinases are inhibited by:
•Pyruvate (if kinase is inhibited then the complex is NOT inhibited)

Kinases are activated by:
ATP, Acetyl CoA and NADH (it will then shut down the complex)

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PDH Complex activity with the cofactors
Lets follow the carbons...

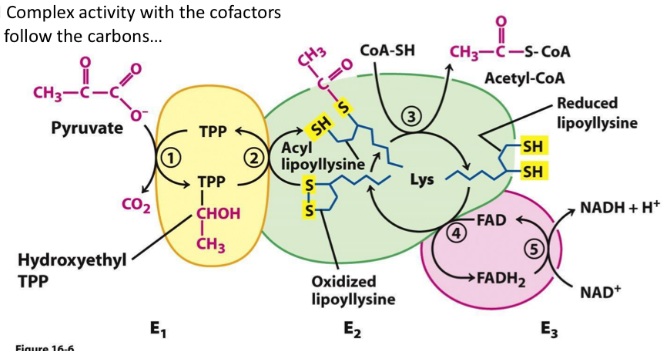

Figure 16-6

Malfunctions of Citric Acid Cycle

Clicker question coming up next

Fluoroacetate inhibits aconitase
substrate for citrate synthase, converted to fluorocitrate which then inhibits aconitase (binds to the Fe^{2+} ion at the active site)

Malonate inhibits Succinate Dehydrogenase
competitive inhibitor due to similarity of structure to succinate

Vitamin deficiencies:
- Niacin (B_3)
- Riboflavin (B_2)
- Thiamine (B_1)
- Pantothenate-B5 (CoA is derived from B5)
- Cobalamin (B_{12}) (required for anapleurotic reaction which generates succinyl CoA from methylmalonyl CoA

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